BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 070 · Procedures and Variable Scope

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What distinguishes a function from a procedure?

Q2. What is SMTP used for?
A. Downloading web pages
B. Sending email
C. Transferring file archives
D. Looking up domain names

Q3. “Cloud storage guarantees that files remain accessible during every Internet outage.” Correct this. Explain why.

Q4. A school buys network devices from different makers. “Standards mean all devices must have the same maker.” Correct this. How do standards help?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Procedure — A named subprogram that performs a task without returning a function value.

Parameter — A named input to a procedure or function.

Local variable — A variable created inside a subprogram, ordinarily limited to its scope.

Global variable — A variable declared for use across program sections, which should be handled carefully.

Algorithm — A defined set of processing instructions.

Testing — Checking expected outcomes against actual program behaviour.

New learning

Procedure: A named subprogram that performs a task without returning a function value. In the welcome message, identify where this technique applies.

Parameter: A named input to a procedure or function. In the ticket confirmation, identify where this technique applies.

Local variable: A variable created inside a subprogram, ordinarily limited to its scope. In the club announcement, identify where this technique applies.

Global variable: A variable declared for use across program sections, which should be handled carefully. In the text output routine, identify where this technique applies.

Worked example

A welcome message uses procedure and parameter. Explain what each contributes. [4 marks]

1. Identify the program's purpose and the pieces of data needed.

2. Procedure [1]; A named subprogram that performs a task without returning a function value. [1].

3. Parameter [1]; A named input to a procedure or function. [1].

4. Each feature needs a correct technical fact plus an explanation.

We Do · Practise with guidance

Guided 1

A ticket confirmation needs procedure. Explain the purpose. [2 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Worked method: Procedure [1]; A named subprogram that performs a task without returning a function value. [1].

Guided 2

Explain parameter for a club announcement. [2 marks]

Worked method: Parameter [1]; A named input to a procedure or function. [1].

Guided 3

When programming a text output routine, explain why local variable and global variable matter. [4 marks]

Worked method: Local variable [1]; A variable created inside a subprogram, ordinarily limited to its scope. [1]. Global variable [1]; A variable declared for use across program sections, which should be handled carefully. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write the call that displays a greeting for "Mia".

```
procedure greet(person)
    print("Hello " + person)
endprocedure
```

write

Question 1

State TWO relevant ideas or operations from this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 2

Explain parameter and local variable in a sports registration. [4 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 3

A programmer makes a library reminder. Explain the roles of procedure, local variable and global variable. [6 marks] Source: Original OCR J277-style teaching question; not an official past paper.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A procedure is a named subprogram that performs a task without returning a value.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A function returns a value to the caller. A procedure performs a task without returning a function value.

Q2. B — Sending email

Q3. Storage, software or processing on remote servers, accessed over a network. Access depends on the connection and service.

Q4. Agreed standards let hardware and software from different makers work together and exchange data.

I Do / We Do pages · Guided practice

Q1. Procedure [1]; A named subprogram that performs a task without returning a function value. [1].

Q2. Parameter [1]; A named input to a procedure or function. [1].

Q3. Local variable [1]; A variable created inside a subprogram, ordinarily limited to its scope. [1]. Global variable [1]; A variable declared for use across program sections, which should be handled carefully. [1].

You Do pages · Code First

```
procedure greet(person)
    print("Hello " + person)
endprocedure
greet("Mia")
```

Expected result: Hello Mia

You Do pages · Independent questions

Q1. Procedure [1]; Parameter [1].

Q2. Parameter [1]; A named input to a procedure or function. [1]. Local variable [1]; A variable created inside a subprogram, ordinarily limited to its scope. [1].

Q3. Procedure [1]; A named subprogram that performs a task without returning a function value. [1]. Local variable [1]; A variable created inside a subprogram,

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

ordinarily limited to its scope. [1]. Global variable [1]; A variable declared for use across program sections, which should be handled carefully. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: